

SIGNALS (OS)

NAME: Guttikonda Gayathri

ROLL NO: 2520030465

1.Signal

Code:

```
gayathri@Gayathri: ~  
GNU nano 8.7.1 Signal.c  
#include <stdio.h>  
#include <signal.h>  
#include <unistd.h>  
  
void handler(int sig)  
{  
    printf("\nSIGINT received!\n");  
    printf("Signal number = %d\n", sig);  
}  
  
int main()  
{  
    signal(SIGINT, handler);  
  
    while (1)  
    {  
        printf("Program is running...\n");  
        sleep(2);  
    }  
  
    return 0;  
}
```

Output:

```
gayathri@Gayathri:~$ nano Signal.c  
gayathri@Gayathri:~$ gcc Signal.c -o Signal  
gayathri@Gayathri:~$ ./Signal  
Program is running...  
Program is running...  
Program is running...  
Program is running...  
^C  
SIGINT received!  
Signal number = 2  
Program is running...  
Program is running...  
^\\Quit  
./Signal  
gayathri@Gayathri:~$
```

2.Sigterm

Code:

```
GNU nano 8.7.1 sigterm.c
#include <stdio.h>
#include <signal.h>
#include <unistd.h>

void handler(int sig)
{
    if (sig == SIGTERM)
        printf("\nSIGTERM received!\n");
}

int main()
{
    signal(SIGTERM, handler);

    printf("Process ID: %d\n", getpid());
    printf("Waiting for SIGTERM...\n");

    while (1)
    {
        sleep(2);
    }

    return 0;
}
```

Output:

```
gayathri@Gayathri:~$ nano sigterm.c
gayathri@Gayathri:~$ gcc sigterm.c -o sigterm
gayathri@Gayathri:~$ ./sigterm
Process ID: 606
Waiting for SIGTERM...

SIGTERM received!
```

3.Sigchild

Code:

```
GNU nano 8.7.1 sigchild.c *
#include <stdio.h>
#include <signal.h>
#include <unistd.h>
#include <sys/wait.h>

void child_handler(int sig)
{
    printf("SIGCHLD received: Child has terminated\n");
    wait(NULL);
}

int main()
{
    signal(SIGCHLD, child_handler);

    pid_t pid = fork();

    if (pid == 0)
    {
        printf("Child is running...\n");
        sleep(2);
        printf("Child is exiting...\n");
    }
    else
    {
        printf("Parent is waiting...\n");
        sleep(5);
        printf("Parent finished.\n");
    }

    return 0;
}
```

Output:

```
gayathri@Gayathri:~$ nano sigterm.c
gayathri@Gayathri:~$ nano sigchild.c
gayathri@Gayathri:~$ gcc sigchild.c -o sigchild
gayathri@Gayathri:~$ ./sigchild
Parent is waiting...
Child is running...
Child is exiting...
SIGCHLD received: Child has terminated
Parent finished.
```

4.Sigpipe

Code:

```
GNU nano 8.7.1 sigpipe.c *
#include <stdio.h>
#include <unistd.h>
#include <signal.h>
#include <stdlib.h>

void handle_sigpipe(int sig)
{
    printf("\nSIGPIPE received!\n");
    printf("The reading end of the pipe is closed.\n");
    exit(1);
}

int main()
{
    int fd[2];

    signal(SIGPIPE, handle_sigpipe);

    pipe(fd);

    printf("Pipe created.\n");

    // Close the reading end
    close(fd[0]);

    printf("Reading end closed.\n");

    printf("Trying to write to the pipe...\n");

    write(fd[1], "Hello", 5);

    printf("Write completed.\n");

    return 0;
}
```

Output:

```
gayathri@Gayathri:~$ nano sigpipe.c
gayathri@Gayathri:~$ gcc sigpipe.c -o sigpipe
gayathri@Gayathri:~$ ./sigpipe
Pipe created.
Reading end closed.
Trying to write to the pipe...

SIGPIPE received!
The reading end of the pipe is closed.
```

5.Sigsegv

Code:

```
GNU nano 8.7.1 sigseg.c *
#include <stdio.h>
#include <signal.h>
#include <stdlib.h>

void handle_sigsegv(int sig)
{
    printf("\nSIGSEGV received!\n");
    printf("Invalid memory access detected.\n");
    exit(1);
}

int main()
{
    signal(SIGSEGV, handle_sigsegv);

    printf("Program started.\n");

    int *p = NULL;

    printf("Trying to access invalid memory...\n");

    *p = 10;

    printf("Program completed.\n");

    return 0;
}
```

Output:

```
gayathri@Gayathri:~$ nano sigseg.c
gayathri@Gayathri:~$ gcc sigseg.c -o sigseg
gayathri@Gayathri:~$ ./sigseg
Program started.
Trying to access invalid memory...

SIGSEGV received!
Invalid memory access detected.
gayathri@Gayathri:~$ |
```